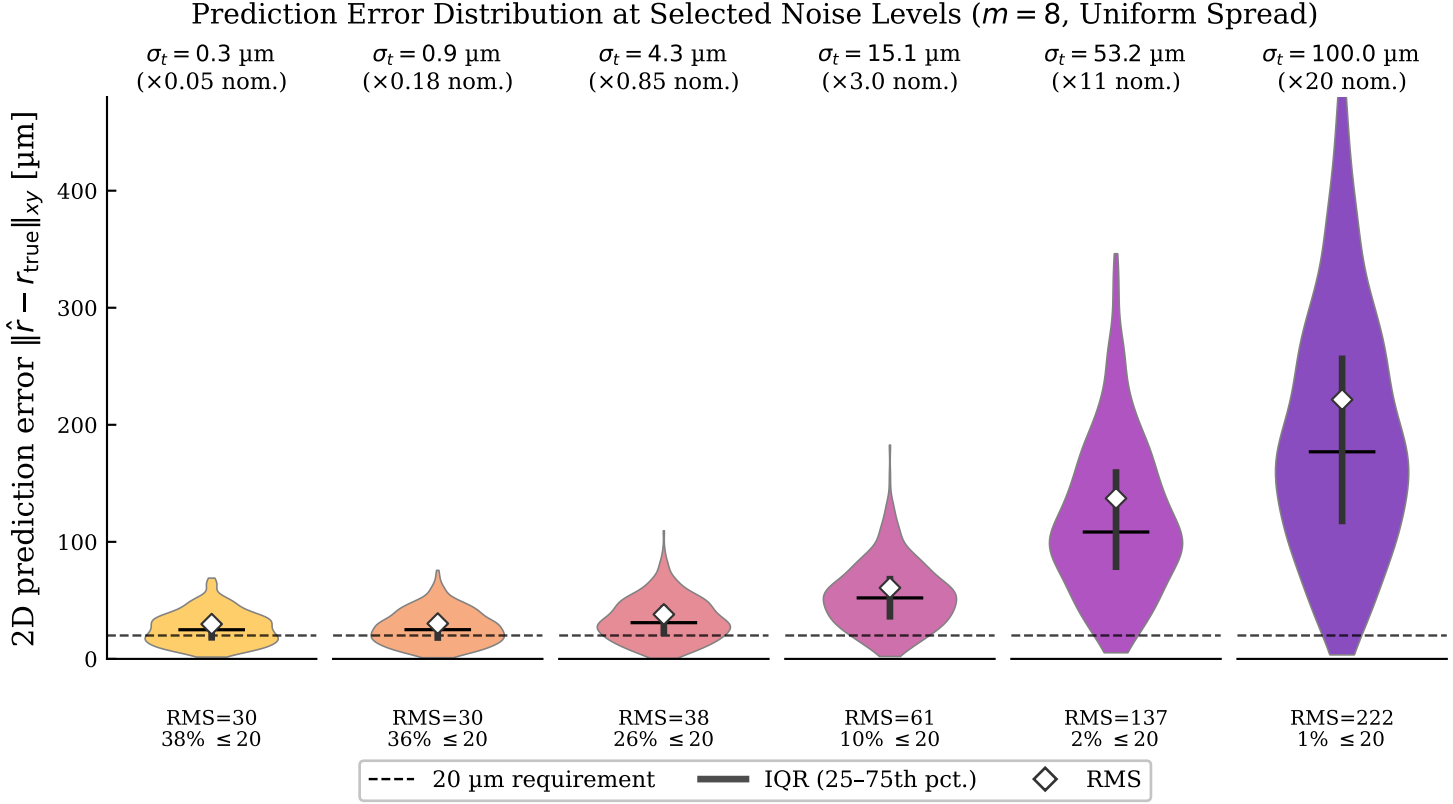
Prediction Error Distribution at Selected Noise Levels ($m = 8$, Uniform Spread)
$\sigma_t = 0.3$ µm (×0.05 nom.)
$\sigma_t = 0.9$ µm (×0.18 nom.)
$\sigma_t = 4.3$ µm (×0.85 nom.)
$\sigma_t = 15.1$ µm (×3.0 nom.)
$\sigma_t = 53.2$ µm (×11 nom.)
$\sigma_t = 100.0$ µm (×20 nom.)
2D prediction error $\|\hat{r} - r_{\text{true}}\|_{xy}$ [µm]
0
100
200
300
400
RMS=30 38% ≤ 20
RMS=30 36% ≤ 20
RMS=38 26% ≤ 20
RMS=61 10% ≤ 20
RMS=137 2% ≤ 20
RMS=222 1% ≤ 20
20 µm requirement
IQR (25–75th pct.)
RMS